

LYCANS AERODESIGN TEAM Trainee Assignment 01 –

Fixed-Wing Competition Research

Name: Trainee Lycans Team

Assignment: Fixed-Wing Competition Research

Date: August 18, 2026

1 Competition Overview

This report outlines my research on the IMechE UAS Challenge 2026 to guide our upcoming fixed-wing and autonomous systems design.

- **Competition Name:** IMechE UAS Challenge 2026
- **Country:** United Kingdom (UK) [A1.2.3]
- **Official Website:** <https://imeche.org/events/challenges/uas-challenge>
- **Short Description:** The IMechE UAS Challenge is an annual educational competition for university students to design, build, and demonstrate an uncrewed air system (UAS) for a specified mission. The 2026 scenario is an agricultural mission requiring autonomous navigation and water payload delivery.[A1.1.1]

2 Important Deadlines and Milestones

The competition runs on a structured engineering timeline with Key deadlines are listed in Table 2.

Table 1: webinars and events[A1.3]

Milestone	Date
Welcome Webinar	12th November 2025
2nd Briefing Webinar	27th February 2026
3rd Briefing Webinar	22nd April 2026
Pre-Flyoff Webinar	10th June 2026
Live Fly-Off Event	29th June – 2nd July 2026

NOTE: Briefing webinars will be provided, on Zoom, by the IMechE and UAS Challenge Volunteers throughout the year.[A8.2.1]

Table 2: Deliverables[D1.2]

Deliverable	Date
Concept Paper Submission	21st November 2025
X-Plane FTR	16th February 2026
Design & Development Spec. (DDS)	20th March 2026
Flight Readiness Review (FRR) Minutes	Prior to Fly-Off
FRR Certificate (Form 701)	12th June 2026

These deliverables will add to the overall competition score as well as some of the individual prizes [D1.3] .

NOTE: For document naming conventions see [D1.1.5 , D1.1.6 , D1.1.7]

3 Entry Fees

A mandatory registration fee of **£950 + VAT** (Pounds Sterling) is payable upon submitting the entry form (Rule 3.3.1) [1]. This fee is non-refundable and contributes directly to the operation of the demonstration airfield; the IMechE does not fund internal aircraft development, fabrication, or team travel expenses (Rule 3.3.1, 3.3.2) [1].

4 Competition Format

The live fly-off event occurs over four days at BMFA Buckminster, UK [Rule 4.2]

- **Deliverables:** Teams must submit a series of deliverables throughout the year, including a Concept Paper, Design Reviews, Simulation Reports, and a Business Presentation.[A1.2.2, Section D]
- **Scoring:** Points are awarded for both documentation and live performance. The total possible score is **1305**, with categories for design, flight performance, innovation, safety, and more. [D1.3]
- **Team and Conduct Requirements:** Teams must follow strict rules on eligibility, conduct, and safety. Only one air vehicle per team is allowed at the fly-off.[A3, A4, F2.2]

5 Manual or Autonomous Control

Teams must design, build, and demonstrate a UAS capable of autonomously navigating while carrying a water payload and dispensing a set amount of water along a defined track between two waypoints. [S2.1.1, S2.3.1, S2.4.1, S2.5.1] Missions consist of :

- **Long Mission (Autonomous):** The aircraft must automatically complete multiple laps of waypoint navigation and perform two separate payload dispersals between waypoints before landing.[S2.4.1] .
- **Short Mission(Manual)** The short mission is a fully autonomous flight requiring the UAS to take off, complete a lap of waypoint navigation, perform a single payload dispersal, and land, focusing on basic autonomous operation and accurate payload release [S2.3.1]

- **Manual Handling:** The manual handling mission is flown by the competition pilot using manual controls, consisting of take-off, a handling assessment, and landing, to evaluate the aircraft's controllability and safety under manual operation [S2.5.1]



Figure 1: The air vehicle shall be capable of automatically navigating around the course defined by waypoint coordinates, which will be provided by the organisers at the final pre-fly off webinar.

6 Aircraft Requirements

Technical parameters and restrictions affecting aircraft design include:

- **Maximum Take-Off Mass (MTOM):** Must not exceed 10.0 kg[T2.2.1]
- **Propulsion:** Electric propulsion only, restricted to standardized **6s 6000 mAh Li-Po batteries** with XT90 connectors [T2.10, T2.1.1, T2.10]
- **Safety Links:** Two highly visible, externally removable XT90 safety loop links are mandatory (Link 1: isolates avionics, Link 2: isolates propulsion ESCs) [T3.2.1, T3.2.2 , T3.2.3]
- **Weather Resilience:** Must operate in winds up to 20 kn (gusts to 25 kn) and light rain [T2.6.1]
- **Contrast:** At least 25% of the surface area must feature highly contrasting colors relative to green, brown, and yellow vegetation [T2.4.5].
- **Storage Box:** Complete system (including aircraft and GCS) must fit inside a box $\leq 1500 \text{ mm} \times 600 \text{ mm} \times 600 \text{ mm}$ [T6.1.1 , T6.1.2 , T6.1.3]
- **Assembly:** Routine toolless assembly from stowed to flight readiness must occur in $\leq 8 \text{ min}$ [F3.4.2]

7 Comparison With Lycans' Competitions

The UAS Challenge is compared side-by-side with our previous competitions, AIAA Design/Build/Fly (DBF) and the ICMTC UAV Challenge, in Table 4 and Figure ??.

Table 3: Cross-Competition Parameters and Engineering Constraints

Parameter	IMechE UAS Challenge 2026	AIAA Design/Build/Fly (DBF)
Mission Focus	Autonomous agricultural spraying	Varying annual cargo missions
Control Format	Autonomous flight (Manual override)	Strictly Manual RC Pilot
Battery Limit	Standardized 6s 6000 mAh Li-Po	Variable chemistry/weight caps
Scoring Strategy	Linear points + Innovation bias	Multiplicative efficiency formula
Payload Type	Liquid water (precise release)	Solid weights / folding structures
Structural Box	Box $\leq 1500 \times 600 \times 600$ mm	Wingspan / folding limits

Table 4: Cross-Competition Parameters and Engineering Constraints

Parameter	IMechE UAS Challenge 2026	ICMTC UAV Challenge
Mission Focus	Autonomous agricultural spraying	Target reconnaissance
Control Format	Autonomous flight (Manual override)	Autonomous waypoint flight
Battery Limit	Standardized 6s 6000 mAh Li-Po	Typically unconstrained
Scoring Strategy	Linear points + Innovation bias	Mission precision + Reviews
Payload Type	Liquid water (precise release)	Gimbal camera payloads
Structural Box	Box $\leq 1500 \times 600 \times 600$ mm	Military-ruggedized standards

References

- [1] Institution of Mechanical Engineers (IMechE), *IMechE UAS Challenge 2026 Competition Rules*, Version 4, released September 12, 2025.
- [2] UAS Challenge Organising Committee, *UAS Challenge 2026 rules released*, IMechE Aerospace Division, 12 September 2025.